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Procedural generation: generating music using a bi-directional two-model system for complex generation

Procedural generation using bi-directional two-model system algorhithm design using an iterative software design method.

Master's thesis in Cybernetics and Robotics
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June 2026

Norwegian University of Science and Technology
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NTNU

ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

PREFACE

Write the preface of your thesis here.

You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

This thesis is written as the final master project at Norwegian University of Science and Technology (NTNU) during the spring of 2026, supervised by Sverre Hendseth.

[1]

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All examples can be found at my NTNU-webpage where they can be listened to, numberings are consistent with the thesis for convenience.

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

AI	Artificial Intelligence
LISP	List Processing
LLM	Large Language Models
MIT	Massachusetts Institute of Technology
NTNU	Norwegian University of Science and Technology
OOP	Object Oriented Programming
PG	Procedural Generation
WFC	Wave Function Collapse
QM	Quantum Mechanics

CHAPTER

ONE

INTRODUCTION

For hundreds of years composers have been creating music using external tools like dice to help them generate something new. As early as the 18th century the idea of using randomness to help the composing process has been used. The musical dice game, "Musikalisches Würfelspiel", was used to randomly generate music based on precomposed parts. [2] This is a very basic way of generating music, however by the random nature of the dice it will not be able to create anything coherent and meaningful like a human composer can. Additionally this process only used precomposed parts which were designed to work together. If we want to generate everything from scratch we need a way more sophisticated system. This is where procedural generation comes into the picute. If we base the meldoy on something random and then use the conventions and "rules" for melodic composition we might be able to generate music that sounds somewhat coherent.

Today the most common way to go about generating music is using Artificial Intelligence (AI) and Large Language Models (LLM). This enables anyone to create music, however these models are trained on large data sets and imitates what has come before. For this thesis the focus is on the craft of creating music, breaking it down to its very primitive form, only pitches and durations. This will enable the user to have more control over exactly what is being generated. However this requires us to create a system and models that capture the essence of how exactly we are to go about contructing the very basic musical ideas. It will not depend on large data sets and will create a musical ideas that can be processed into a fully fledged musical idea.

If we are able to solve this problem of creating a good structure and model for this kind of generation we will not only have a solution for musical ideas, but this kind of system can solve many kinds of problems.

The problem we want to solve is the problem of creating a coherent system that will be able to generate music that adheres to certain musical principles. In order to do this we need to consider many things, and system design is a big part of this. When buidling such a program we need to consider what parts it should consist of as well as how they are supposed to interact. Object Oriented Programming (OOP) is a solid way to approach this system design.

The solution will be a system that is modular, easy to modify and generate new music using newly defined constraints or inputs on structure, motifs etc...

CHAPTER

TWO

BACKGROUND

This chapter provides the background theory for the project. First we give an overview of procedural generation and some of the different methods used in procedural generation. Then we give an overview of some basic music theory that is necessary to fully grasp the structure of the system. Lastly we cover some about object oriented programming.

2.1 Procedural generation

Procedural Generation (PG) is algorithmic generation of content. This content can take many forms including landscape, levels, textures or even 3D models. This leads to smaller file sizes since the program can generate the content using algorithms rather than having everything pre-made and saved.

This section will cover several methods of PG and discuss similarities, differences and most importantly limitations of the methods. Additionally we will take a look at and what we are trying to achieve with the procedural generation in this project and propose a new method that will handle the problem of music generation in a new way.

2.1.1 Wave function collapse

Wave Function Collapse (WFC) is a method used to procedurally generate using either given constraints, or it can also use some input that it analyses and generates the constraints procedurally which are then used to generate something in the same style as the input.

It works by initializing the entire space in a superposition which means that the state is yet to be determined. We call this an *unobserved* state, meaning we don't know what state it is in yet. These unobserved states are not complex numbers like they are in Quantum Mechanics (QM), but the process is similar which is why it gets its name inspired by the field of QM. Randomly assigning parts of the space we are able to collapse[1] some of the *wave function*[2]. In Figure 2.1.1 we

[1] collapsing a superposition means that we decide the actual state of the given position in the space

[2] the wave function is simply the constraints what decide what the state of the object should be dependent on external factors

Figure 2.1.1: Illustration of how procedural tiles from input image is used to generate a new image. [3]

can see an illustration of how the "tiles" of an input image re-appear in the final image generation.

2.1.2 PG using constraint satisfaction

Similar to WFC we have PG using constraint satisfaction. This method uses given constraints for influencing the generation. It usually works by randomly generating something in the given domain, then iterates over the generated data and enforces constraints in order to make it adhere to a model designed by the programmer or end user. After several iteration over the domain we end up with a domain where the constraints are satisfied for all the objects in the domain.

2.1.3 L-systems (Lindenmayer systems)

Lindenmayer systems are a method of PG based around the idea of creating rules for the generation which lead to self-similarity and fractal-like forms. This kind of PG is often used in generating organic structures such as trees or other plant-like structures.

2.1.4 Binary space partitioning

By partitioning the given space in smaller spaces recursively we are able to find a subdivision small enough to generate something that propagates to the higher space with certain characteristics.

2.1.5 Random noise and seeds

Perlin noise is a very common way to generate structures like worlds etc... It is very good at creating smooth transitions from one state to another, like height maps.

2.1.6 Random walks

Random walks are often used in generating maps or other spaces. The random walk works by having an entity walk around "drunkenly", in this case meaning randomly. Then the path is traced, either to create a circumference of an area or

the traversed spaces become the area itself. Oftentimes the drunk walker has a set amount of time or steps it can walk before it is terminated, then a new walker is placed at a random location, this helps to smooth out the generated area.

The same technique can be used to create textures as well, in REFERENCE HERE we can see how changing the colours and directional bias we can generate textures that imitate real life. In the tree and sand example the directional bias is vertial for the tree texture and diagonal for the sand texture. For the map texture it is uniform directional bias, but colouring to indicate water, land and mountains. The least traveled to square get more blue, and the more frequent it goes to green and brown/white to indicate mountains.

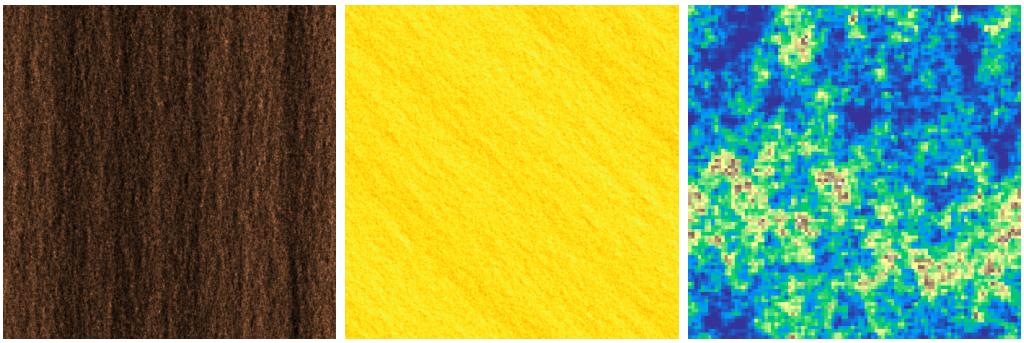


Figure 2.1.2: Three different textures generated using random walks.

2.2 Music theory

This section contains some basic music theory relevant to understand how music is constructed. It is not neccessary to understand everything in depth, but farmiliarity with these concepts will be of great help to understand the project as a whole. Throughout the thesis there will be several examples and figures using musical notation so a basic understanding of how to read these figures will be benefitial.

2.2.1 Notes

The very core of music is composed of notes. Notes are vibration in air which we interpret as pitches. In the western system we agree upon the standard of pitches is based around A_4 which is $440Hz$. All other pitches are tuned in relation to this reference.

2.2.2 Note values

The same musical pitch can be written in several different ways in order to dictate the duration of the note. This is done by using variations of the note. This can be by Refering to Figure 2.2.1 we can see the different parts of the note highlighted with colour. The note-head is coloured red and the placement of this decides which note is represented. The stem can go either up or down, this does not change the meaning it just makes it more compact to make it go down if the note placement is high on the staff. The direction of the stem can carry meaning if there are two notes simultaneously on a staff, then it usually indicates two different voices. Meaning one player playes the melody with stem up and one with the stem down. Next we have the beams and the flags coloured in purple and blue respectively. These two carry the same meaning, one flag is the same as one beam, meaning they have the same duration. The beam simply makes it less visually crowded when reading. The omitting of beams and only using flags *can* mean that the notes are more separated, but this is not neccessarily the case.

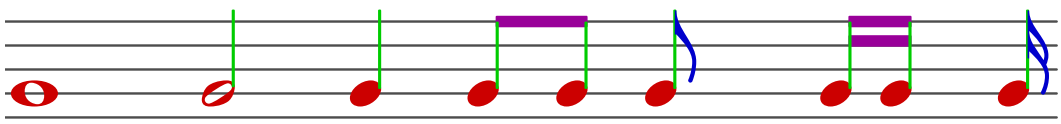


Figure 2.2.1: The figure shows the parts of the musical note. The note-head in red, stem in green, beam in purple and flag in blue.

2.2.3 Scales

The musical scale is building block for creating not only melodies but also harmony, which we will cover a bit later. Firstly the musical scale is, for the most part, constructed using seven notes. In the western canon this is usually seven

Table 2.2.1: Table with common note values, their names and their corresponding duration.

Symbol	Name	Rest symbol	Duration
𝅝	Whole note	𝄻	4 beats
𝅗𝅥	Half note	𝄼	2 beats
𝅘𝅥	Quarter note	𝄽	1 beat
𝅘𝅥𝅮	Eigth note	𝄾	1/2 beat
𝅘𝅥𝅯	Sixthteenth note	𝄿	1/4 beat
𝅘𝅥𝅰	Thirtysecond note	𝅀	1/8 beat

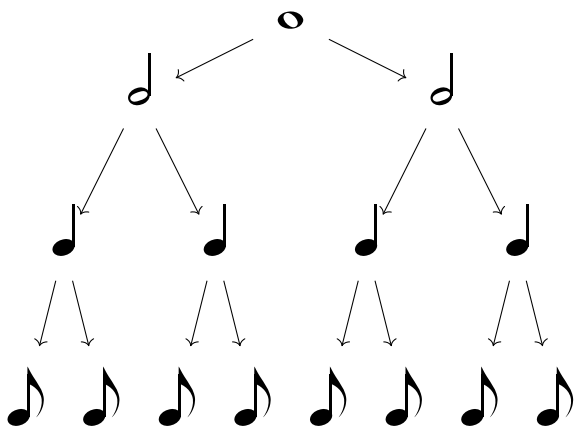


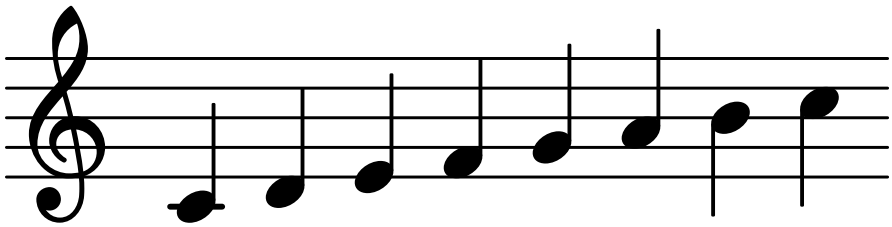
Figure 2.2.2: Tree sctructure showing how you can divide the whole note into the smaller note values. Here each level in the tree takes the same amount of time, meaning the whole note at the top takes as long as the eight eigth-notes at the bottom.

notes. You can see the basic c-major scale in Figure 2.2.1, it is comprised of the notes c d e f g a b c. The tones of the scale can be named according to their function within the scale. We often label the notes $\hat{1}$, $\hat{2}$, $\hat{3}$, $\hat{4}$, $\hat{5}$, $\hat{6}$, $\hat{7}$, and then starting back at $\hat{1}$. The use of the carets are called the Nashville Number System, this is used to differentiate the scale degrees from the chord tones which will be discussed in Section 2.2.7 When we consider chord structures we use the numbers but usually count above from 8 to signify harmonic functions, but when considering only scales and melodies we usually only use 1 through 7

The most important function is that of the scale degree 1, this is the *tonic*, which is the 'home' of the key. This is the most stable function there is, it feels resolved and does not have any tension urging it to move anywhere.

Next up we have the dominant function, there are two notes degrees which fall into this category, that is the 5th and 7th scale degree. These notes have tension. They want to move, especially the 7th scale degree, it has a great pull toward the tonic. You can hear this in the example under Example 2.2.1. The penultimate tone is the 7th scale degree and it leads to the tonic.

To lead us to the tonic functions we have what are called the pre-dominant function. This simply means 'before dominant', and their function is to set up the tension to lead us to the dominant degrees which in turn leads us back to the tonic.



Example 2.2.1: The C-major scale written in the treble clef.

Example 2.2.2: The E♭-major scale written in the treble clef.

2.2.4 Keys

The concept of a musical key is simply shifting around where the tonic note is. For instance in Example 2.2.1 you can see the major scale in C. And in Example 2.2.2 you can see the major scale in E♭. In todays tuning system there are no differences between the keys, meaning that the sound the same, just higher and lower. This was not the case earlier in history, where the temperament(how the tones were tunes in relation to each other) of the keys differed. Today it just sounds higher or lower depending on the key.

2.2.5 Melodies

Melodies is the horizontal component of music, meaning we look at it as a time-series. We have a given note at a given length followed by another note or rest of a given length etc... This means that you can look at a melody as a series of intervals as well. If we do this on known melodies we will observe that they are mainly composed of steps and skips. This means that the distance between one note and the next is one or two tones. Large leaps are usually leaping to important tones like the tonic or the root of the current chord.

2.2.6 Intervals

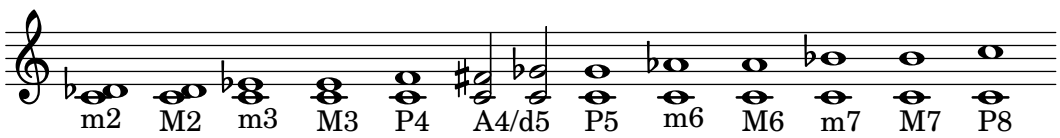
Before discussing harmony we need to consider the different types of distances between notes. The musical distance between two notes are called an *interval*.

Table 2.2.2: Table containing all musical intervals up to an octave, the number of semitones and their abbreviated form.

Interval name	Semitones	Abbreviation
Unison	0	P1
Minor second	1	P1
Major second	2	m2
Minor third	3	m3
Major third	4	M3
Perfect fourth	5	P4
Tritone[3]	6	TT or aug4/dim5
Perfect fifth	7	P5
Minor sixth	8	m6
Major sixth	9	M6
Minor seventh	10	m7
Major seventh	11	M7
Octave	12	P8

Intervals can be identified by the number of semitones between the notes. Additionally we have special names for all the intervals, these names can be seen in Table 2.2.2. The list also provides the number of semitones between the notes.

In Figure 2.2.3 we can see how all the normal intervals look on a stave. Naming the interval on a general basis means looks at the distance, this means that a second is two consecutive notes and a third has "one space" between the notes, this is regardless whether it is minor or major a third looks the same distance wise regardless of quality. You can see this in Example 2.2.3, here we observe that the distance between the notes in a minor third and a major third is the same, the only difference is that the minor third is one semitone smaller, but that is shown by the ♭-symbol which means the note has been lowered by a semitone. Consequently for the tritone we see there are two ways to spell the same interval, one is a fourth that has been raised and the other is a fifth that has been lowered. These two are *enharmonically equivilant* which just means that they are the same pitches, they are just spelled differently. For the augmented fourth(the first spelling of the tritone) we se the ♯-symbol used, which simply means the note is raised by a semitone.



Example 2.2.3: All the musical intervals from the unison to the octave including their variants.

[3]can also be called diminished fifth or augmented fourth, where diminished simply means lowered by semitone and augmented means raised by one semitone

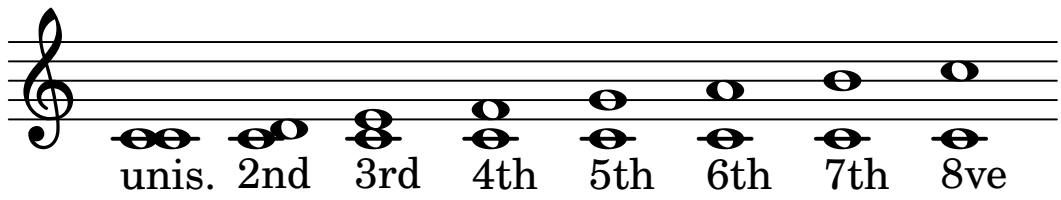


Figure 2.2.3: All normal intervals from the note C_4 devoid their variants in quality.

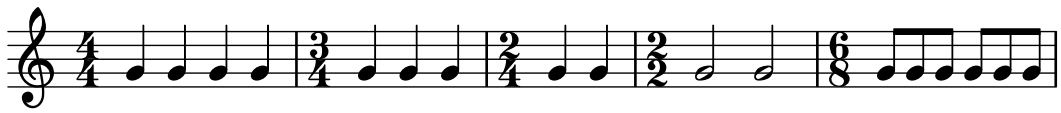


Figure 2.2.5: Demonstration of a few common time signatures.

2.2.7 Harmony

Harmony is the vertical component of the music, harmony is what happens when two or more notes happen at the same time. This is what chords are build around. When considering the harmonic progression in tonal music we can consider all the standard chords in the major key. They are labeled by numbers using the roman numerals. Upper case means the chord in major and lower case means minor. You can see the difference between major and minor chords in figure [make a figure for this], and listen to the difference at the link provided in the caption.

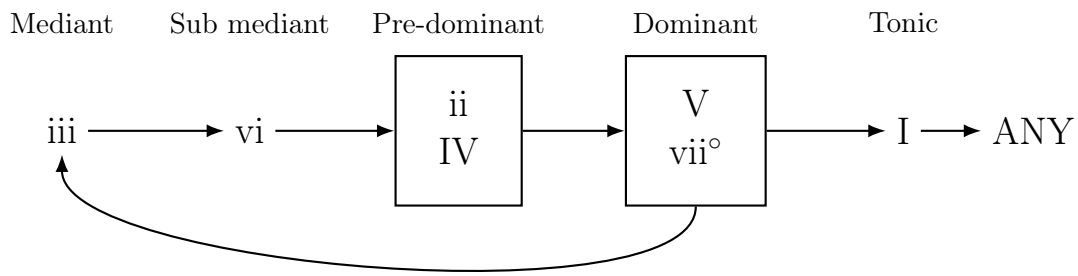


Figure 2.2.4: Harmonic progression in tonal music. The arrows indicate the typical flow of harmonic functions.

2.2.8 Time signatures

The time signature explains what is in a bar of music. A bar is the space between the vertical lines on the staff. A 4/4, sometimes written as $\mathbf{C}$, time means that there are four quarter notes per bar and consequently 3/4 time means that there are three quarter notes per bar.

2.2.9 Musical form

Now that we have an idea of the musical structures on the melodic and harmonic layer we will take a look at musical forms. This is a very important idea when

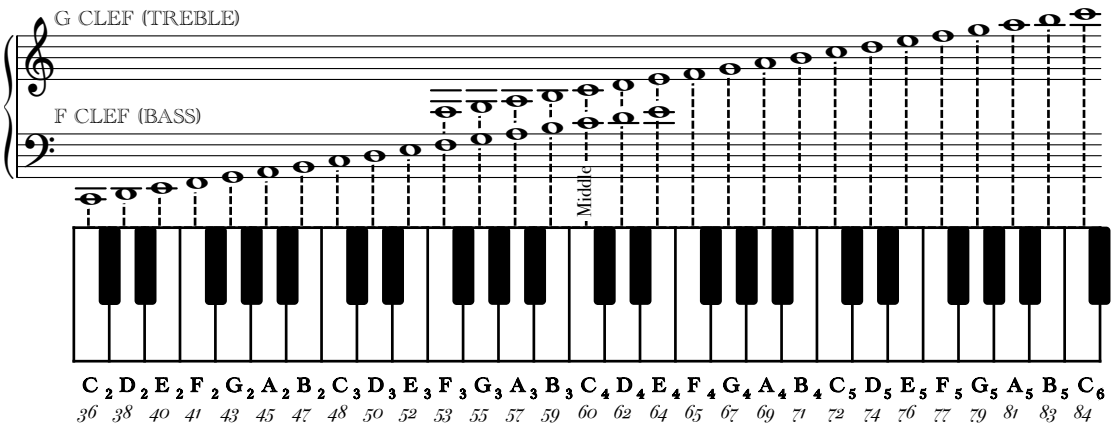


Figure 2.2.7: Overview of note placement on the staff and piano, additionally the corresponding midi numbers are written below the note names. Notice how the number increases by two when there is a black key between consecutive notes.

it comes to constructing melodies. A melody is not an infinitely wandering idea of music. A good melody follows a form. Often this can be viewed as repetitions and variations of a given musical idea. Two very common forms of musical form are the *sentence form* and the *period form*.

In Figure 2.2.6 we can see an example of the musical sentence. It starts with a 2-bar opnening phrase, labeled as the *basic idea*, then we have a repetition of the same idea only adapted for a new chord. Then the following 2 bars are a fragmentation of the idea. Then for the last two bars we have a conclusion, this ties the idea together. The cadence at the end can be considered the end of a sentence. It concludes the idea before it moves on to something else.

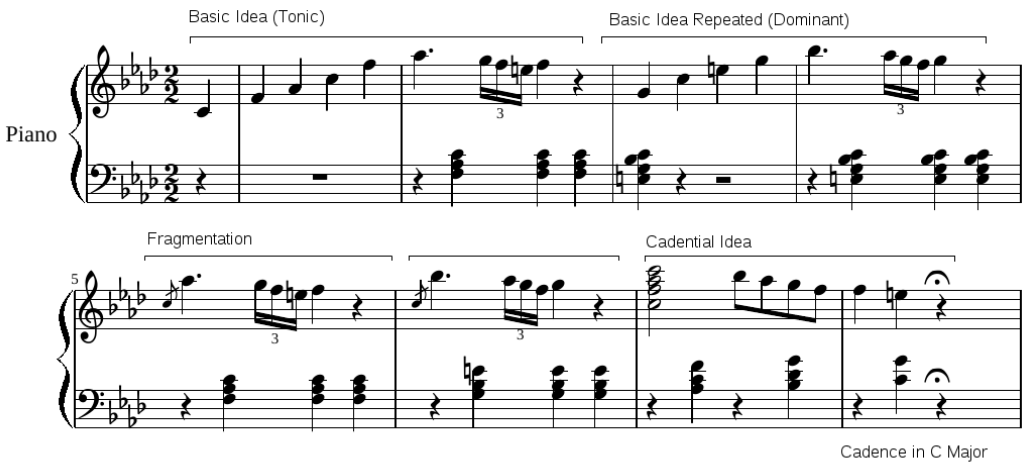


Figure 2.2.6: A classical example of the musical sentence form. These are the opening bars of Bethovens piano sonata in F-minor. [4]

2.3 Object oriented programming

OOP had its inception in the sixties with the *aritficial intelligence* group at Massachusetts Institute of Technology (MIT). Here the term "object" was used in the language family List Processing (LISP) to refer to LISP atoms[4] with with certain attributes or properties. [5]

2.3.1 Features of OOP

Some of the useful features of OOP are that we can achieve using OOP is ...

[4] a LISP atom is an indivisible entity in the language. LISP consist of two data types, atoms and lists. And a list can only contain atoms or other lists

2.4 Iterative development

> A major problem in the design of a computer system is that many aspects of system behaviour and performance are not discovered until the system has been built...[6]
>
> *F.W. Zurcher*
> *B. Randell*

This project utilizes a from of development called iterative design. This means that I do the project by creating a design, taking notes on what needs to be improved, then starting again with the new knowledge gained in the previous iteration.

This sounds like more work, but in reality it leads to an effective workflow. It allows for fast prototyping since we can start programming something straight away, then when the project halts due to the system that has been created is incompatible with the ideas for improving it we take notes on what worked, what is missing, and use this to create a specification for the next iteration.

By doing this we maintain active development which doesn't stagnate and we can always create a new iteration when development slows down. After 3-4 iterations we should be able to land on a system that works in a way that we are happy with.

CHAPTER

THREE

SOFTWARE DESIGN

Include the complete description of the methods used in your research here.

3.1 Object oriented programming

This project will use the widely known process of object oriented programming. In simple terms this simply means that all the parts of the code are structured as objects, meaning everything is represented as objects with given properties. This means that for instance a 'melody object' could consist of all the notes and durations it consists of, the key it is written in and any metadata explaining what the melody represents. Meaning if it is a verse or chorus etc...

In order to design an object oriented program the design process is very important. Creating good models for the objects are imperative to make it work. If the level of abstraction is too high, the sematic gap between the real-world concepts they represent and the acutal implementation in the code will be too large and the model will be un-intuitive and hard to both use and also understand.

CHAPTER

FOUR

IMPLEMENTATION

The implementation of this program started by just writing something that generated something. This was done to not get stuck in the planning phase. You could consider this a contradiction to the part in software design which states that good planning is imperative. However, trying out some code and figuring out what works, what doesn't and what needs to be implemented in the models is very helpful. Therefore the first iteration was a test to get some systems working and finding out what it neccesary to make it work and what is not imporatant.

4.1 The iteration process

After the first iteration lots of issues were observed and were put into the spec for the second iteration which improved upon the model in various ways. Some imporatnt features were restricting the range so that an acutal instrument can play it, be that an instrument or the human voice.

CHAPTER

FIVE

RESULTS

CHAPTER

SIX

ITERATIVE DEVELOPMENT

In this project I have utilized a from of development called iterative design. This means that I do the project by creating a design, taking notes on what needs to be improved, then starting again with the new knowledge gained in the previos iteration.

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By doing this we maintain active development which doesn't stagnate and we can always create a new iteration when development slows down. After 3-4 iterations we should be able to land on a system that works in a way that we are happy with.

6.1 First iteration

The goal of the first iteration was to get a "proof of concept" working. I wanted to test out how I could generate notes and visualize them as well as listen to audio of the notes generated. For this I used *python* to generate *liypond* files. From these files I was able to generate PDFs of sheet music as well as *wave-files* to listen to how it sounds without having to play/sing it myself.

6.1.1 Implementation

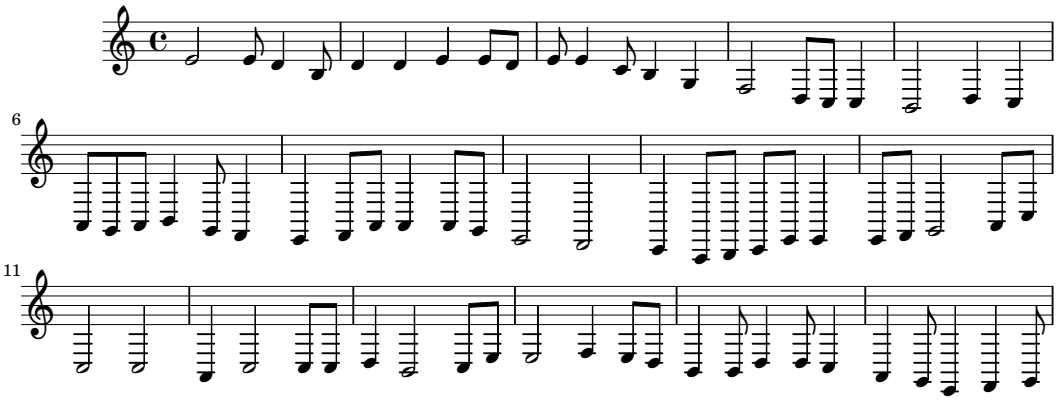
This first iteration was a very primitive handling of generating music. It had very few constraints on the melody and contained little to no metadata on the music which made it unsuitable for actual music.

The code generated the notes within a given scale and restricted the distance (called interval in music theory) to only 2nds and 3rds. However this meant that the melody meandered without any other restrictions. If the generated music was long it would have a tendency to go beyond the range of an instrument or the human voice.

6.1.2 Results

The melody generation works fine for smaller melodies, however there is no self-similarity and no rests. This means that the melodies we generate are not particularly musical or interesting, but it is a great starting point to getting to a fully fledged melody generator. In Example 6.1.3 we can see how a short melody using the first iteration looks.

In Example 6.1.1 you can see what happens when the generated melody is longer, it will tend to go beyond what an instrument will be able to play. You can see this clearly in *bar 6 through 16* (the entire 2nd and 3rd system), where there are lots of ledger lines. Some of these notes are not even available on the grand piano.



Example 6.1.1: Melody showing how the octave is not restricted and ends up in unplayable range.

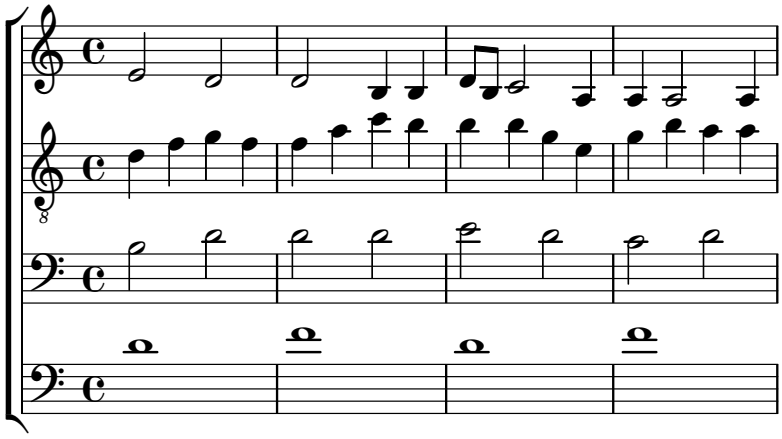
However a good feature that was included was the ability to generate more voices at the same time. This feature was quite primitive since the voices were generated independently and did not adhere to any harmony, however it can be a good skeleton to use later if we get to the point that we want to actually harmonize with the melody we generate. In Example 6.1.2 you can see how we generated four voices all with their own rhythmic restriction. The bottom voice only has whole notes, the second lowest one has only half-notes, the second highest has only quarter and the top voice has a combination of several types.

6.1.3 Discussion

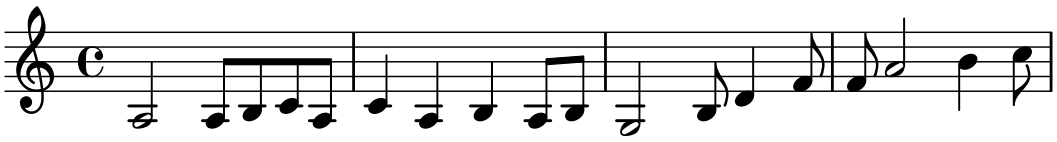
As we can see the first iteration shows a promising proof of concept, however it is quite limited and needs lots of improvement before we have a good system for melody generation.

For the second iteration we need to be able to enforce more restictions on the melody so that it follows more convetions of good melody writing. Additionally we need to have some self-similarity so that the music follows some kind of musical form in order for it to feel coherent and part of the same idea.

The second iteration will focus on creating a generation algorithm that tat is able to generate a melody with a structure. Meaning when generating a melody we will generate a phrase, that phrase should contain an idea, a repetition, and



Example 6.1.2: Melody showing how we can generate multiple voices with different rhythmic restrictions.



Example 6.1.3: Example melody of the first iteration melody generator.

the end of the melody should end resolved with a long note on the tonic of the key. Additionally we need to restirct the range of the melody so that it will be playable/singable.

6.2 Second iteration

CHAPTER

SEVEN

DISCUSSION

Discuss your results here.

The generated music is good because this:

The generated music is bad because this:

etc etc...

7.1 Future work

The current implementation is not complete and there are several features which would benefit the system.

7.1.1 Non-block harmony

Not sure how far the project will go, but probably no implementatino of Non-block harmony happened.

CHAPTER

EIGHT

CONCLUSIONS

Short summary of what the solution does and why it is important that it was made. Also it is hard problem and the solution is doing a good, but not complete job.
Looking at this thing here, [7] [8] [9] it is a citation.

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APPENDICES

A - GITHUB REPOSITORY

All code and latex-files used in this document are included in the Github repository linked below. Further explanations are given in the readme-file.

Github repository link

- https://github.com/lima98/TTK4900

B - AUDIO EXAMPLES

Audio-files for the examples in the thesis can be played from my NTNU-webpage.

Audio example archive

- https://www.janoivil.folk.ntnu.no

C - CODE DOCUMENTATION

All documentation for the code is included in the Github repository linked in 8 it is also provided as a web-page hosted at my NTNU-webpage.

Code documentation

- https://janoivil.folk.ntnu.no/docs/_build/html/index.html